



DPUSocket

1. Static parameters:

<i>Socket_IsTCP/IP</i>	Determines if socket should communicated via TCP/IP [=true] or via UDP [=false].
<i>Socket_IP</i>	Target IP address for the sending part of socket
<i>Socket_IsServer</i>	Determines if the (TCP/IP-) socket should act as a server or as a client.
<i>Socket_PortSend</i>	Port number for the packet to be sent
<i>SendDefinition</i>	Packet definition for the packet to be sent
<i>Socket_PortReceive</i>	Port number for the packet to be received
<i>ReceiveDefinition</i>	Packet definition for the packet to be received
<i>BigEndian</i>	[true/false] Defines how the data transfer should take place.

2. Outputs:

<i>Socket_PackageInSize</i>	Packet size of the packet to be received in Byte
<i>Socket_PackageOutSize</i>	Packet size of the packet to be sent in Byte
<i>Socket_PackageNumberSend</i>	Number of sent packets (While in server mode, only the packets sent to the first connected socket are counted.)
<i>Socket_PackageNumberReceived</i>	Number of received packets

3. Functional principle:

DPUSocket establishes a TCP/IP respectively UDP connection to another computer.

While the simulation is running, it exchanges one data packet with the remote computer per simulation step. A simple binary protocol is used for the data packets. The content of the data packets is defined in the following form:

```
<PacketDefinition> =  
(  
    (,)(<VariableNames> ,<Datatype> [,<Defaultvalue> , <Multiplier> ,  
    <Offset>] )  
) ;
```

The variable names declared in the definition are provided as inputs respectively outputs of the DPUSocket. The supported data types are listed in the following table:

Data type	Data length	Min	Max
byte	1	-128	+127
ubyte	1	0	+255



word	2	-32768	+32767
uword	2	0	+65535
long	4	-2147483648	+2147483647
ulong	4	0	+4294967295
longlong	8	-2^{63}	$+2^{63}-1$
ulonglong	8	0	$+2^{64}-1$
float	4	3.4e +/- 38 (7 digits)	
double	8	1.7e +/- 308 (15 digits)	
string(n)	n	text value containing <i>n</i> bytes of text	

The data packet starts with the first entry. The byte order is determined by the parameter *BigEndian*.

4. Example:

Suppose we wanted to send and receive Messages which consist of a *Type* number and a short *Message*. We could use a script like the following:

```
DPUSocket comm
{
    Index = 10;
    Computer = {LOCALHOST};

    Socket_IsTCPIP = false; // udp-socket
    Socket_IP = "DESIRED_IP";
    Socket_PortSend = 9401;
    Socket_PortReceive = 9402;

    ReceiveDefinition =
    (
        (Type_r, ubyte, 0),           // 0 = Agree, 1= Disagree
                                     // 2= Question, 3= Answer
        (Message_r, string(64))      // The actual message
    );

    SendDefinition =
    (
        (Type_s, ubyte,2),           // 0 = Agree, 1= Disagree
                                     // 2= Question, 3= Answer
        (Message_s, string(64))      // The actual message
    );
};
```

Note: *Socket_IP* holds the IP of the target computer.